

NI USB-6009 Logger — User Manual

Software version: 1.1.0 · **Platform:** Windows 10/11 (64-bit) · **Language:** English (Nederlandse versie: [USER-MANUAL.nl.md](#))

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1. What this software does

The NI USB-6009 Logger turns a National Instruments **USB-6009** data-acquisition device (the "DAQ") into a data logger for a test stand. It can:

- **Log** analog voltages (AI) and, optionally, digital inputs (DI) to a **CSV** or **Excel (.xlsx)** file, with a live graph while it runs.
- **Calibrate**: show a steady, averaged number per channel on screen, without writing any file — for zeroing and checking sensors before a test.
- **Ignite**: run a safe, step-by-step ignition sequence (buzzer warning, igniter continuity and leak checks, hold-to-fire, current confirmation) while logging the whole event.
- **Protect your data**: every logging or ignition run is also written to a *recovery copy* on disk, so a crash, power loss or unplugged cable does not lose what was already measured.

There are two programs sharing the same engine:

Program	For	Start from
NI USB-6009 Logger (GUI)	Normal operation by the operator	Start menu / desktop icon

Program	For	Start from
ni_usb6009_logger (command line)	Scripting, advanced users	A terminal — see chapter 14

This manual concentrates on the graphical program.

2. What you need

Item	Notes
NI USB-6009	With its USB cable.
A Windows PC	Windows 10 or 11, 64-bit, with a free USB port.
NI-DAQmx driver	Free from NI. The installer can include it (see chapter 5).
Installer	NI6009Logger_Setup_1.1.0.exe (or newer).
Sensors / signals	Wired to the USB-6009 as described in chapter 4.
<i>For ignition only:</i> relay board, buzzer, 1 Ω shunt resistor, 47 k Ω resistor, ignition power supply, fuse	See chapter 11 and the wiring diagram in 4.4.

Disk space: the NI-DAQmx driver alone is several hundred MB. The program itself is small.

3. Safety first

Read this before wiring anything, especially if you will use the ignition function.

- **The USB-6009 must never carry igniter current.** It only *measures* the small voltage across the shunt resistor and *switches a relay* through a driver.
- Always drive a relay coil through a **transistor or driver chip (e.g. ULN2803)**, never straight from a DAQ output pin. Put a **flyback diode** across the coil.
- **Fuse** the ignition power supply appropriately.
- Keep the ignition wiring physically **separate** from the DAQ wiring where possible, or use isolation (opto-coupler or an isolated current-sense amplifier).
- **Do a dry run first**, with a dummy load (e.g. a resistor or a lamp) instead of a real igniter, and with the igniter **disconnected from the motor**.
- Nobody may be near the test article while the buzzer sounds or while FIRE is possible.
- If anything looks wrong: press **ABORT**, or simply close the program — both force the output lines LOW (relay off, buzzer off).

4. Connecting the hardware

4.1 The USB-6009 terminals

The device has 32 screw terminals (two rows of 16, behind removable covers). Numbers and names are printed on the device label.

Terminal	Signal	Terminal	Signal
1	GND	17	P0.0 (digital)
2	AI 0 (+)	18	P0.1
3	AI 4 (AI 0 – in differential)	19	P0.2
4	GND	20	P0.3
5	AI 1 (+)	21	P0.4
6	AI 5 (AI 1 – in differential)	22	P0.5
7	GND	23	P0.6
8	AI 2 (+)	24	P0.7
9	AI 6 (AI 2 – in differential)	25	P1.0
10	GND	26	P1.1
11	AI 3 (+)	27	P1.2
12	AI 7 (AI 3 – in differential)	28	P1.3
13	GND	29	PFI 0
14	AO 0	30	+2.5 V
15	AO 1	31	+5 V
16	GND	32	GND

(Always check against the label on your own device.) This software uses the **analog inputs** (AI 0–7), the **digital lines** (port 0: lines 0–7, port 1: lines 0–3). The analog outputs are not used.

Naming used in the program:

On the device	In the program
AI 0 ... AI 7	ai0 ... ai7
P0.0 ... P0.7	port0/line0 ... port0/line7
P1.0 ... P1.3	port1/line0 ... port1/line3

4.2 Connect the device to the PC

1. **Install the software first** (chapter 5) so the driver is ready, then plug in the USB-6009. (Plugging it in first also works; Windows just installs the driver afterwards.)
2. Connect the USB cable directly to a PC USB port — avoid unpowered hubs and long extension cables.
3. The device's LED comes on/blinks when it is recognised. Wait a few seconds.
4. The program recognises the device by itself, normally as **Dev1**. If you ever need to check or rename it, use *NI MAX* (Measurement & Automation Explorer, installed with the driver) → *Devices and Interfaces*.

4.3 Wiring the analog inputs

Choose one of two methods per measurement (the same setting applies to all logged channels — it is the **Term config** field):

RSE — single-ended (default). Each signal needs two wires: signal → AI n , and signal ground → any GND terminal. Works on all eight channels (ai0–ai7). Use for sensors that share a ground with the DAQ; simplest wiring.

DIFF — differential. Each signal uses a *pair* of terminals: + to AI_n, - to the partner terminal (see the table: ai0 uses terminals 2 and 3, ai1 uses 5 and 6, ai2 uses 8 and 9, ai3 uses 11 and 12). Better noise rejection for small or floating signals. **Only ai0–ai3** are available in DIFF mode, because ai4–ai7 are the partner pins. Don't use an ai4–ai7 channel at the same time as the DIFF channel that uses its terminal.

Input range is ±10 V (maximum). **Never apply more than ±10 V** (relative to GND) or the input may be damaged. For signals bigger than that, use a voltage divider.

The USB-6009 has **no** "NRSE" mode, even though the command line accepts the word. Use RSE or DIFF.

4.4 Wiring digital inputs (optional)

Connect a digital signal (0 V = low, up to 5 V = high) between P0.x (or P1.x) and GND. Digital inputs are **static**: the program reads them once per data block (chunk) and repeats that value on all rows of the block. Lower the *Chunk size* to sample them more often (see chapter 8).

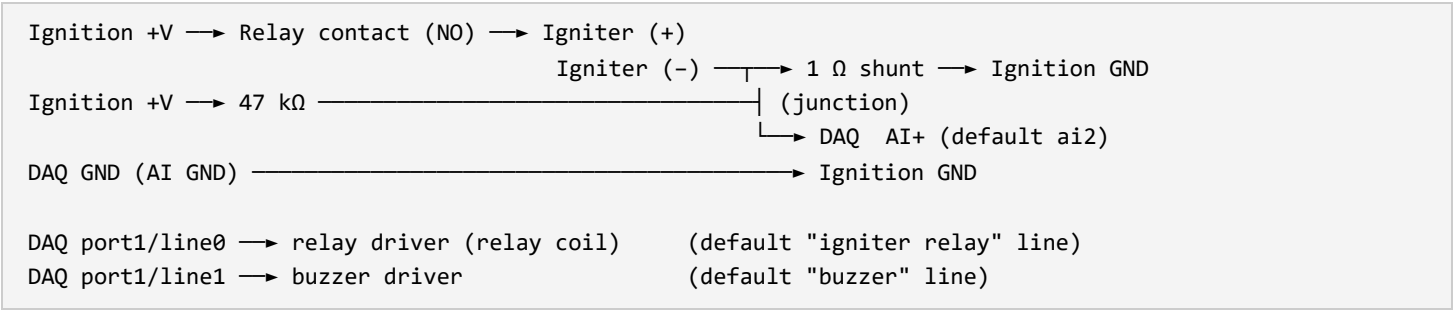
4.5 Wiring the ignition circuit (only for the Ignite function)

This section is the reference wiring used by the software's default settings.

Parts

- **Relay** (via a driver/transistor such as a ULN2803 or MOSFET, with a flyback diode) to switch the igniter. The relay *contacts* carry the ignition current.
- **Buzzer** (driven from a DAQ digital output through a transistor if it needs more than the DAQ's few mA).
- **Shunt resistor**: 1.0 Ω, at least 2 W, low inductance, in the igniter's ground path.
- **Bias resistor**: 47 kΩ, from ignition +V to the igniter/shunt junction. It gives a tiny test current (~0.25 mA at 12 V) used to check that the igniter is connected.

Connections



How the checks work

Relay	Current flowing	Voltage across 1 Ω	Used for
Open	only the bias current (~0.25 mA at 12 V)	~0.25 mV	Continuity check — igniter is connected
Closed	full ignition current (3–6 A typical)	3–6 V	Fire confirmation

The software calculates current = measured voltage ÷ shunt resistance, and compares it to three limits (set on the Ignite tab, see chapter 11).

Mandatory: include the current-sense channel (for example ai2) in the list of *AI channels* on the left side of the window. See chapter 11.

With the example wiring above, set **Term config** and **Sense term config** to **RSE**. If you wire the shunt differentially, use DIFF and a channel in ai0–ai3.

4.6 Checklist before powering up

- ☐ Every signal wire is on the intended terminal, and no wire touches another.
- ☐ No input exceeds $\pm 10\text{ V}$.
- ☐ DAQ GND and ignition GND are connected as in the diagram.
- ☐ The ignition supply is **off** (and fused) while you set things up.
- ☐ The igniter is a dummy load / not installed in a motor for the first test.

5. Installing the software

5.1 Run the installer

1. Double-click `NI6009Logger_Setup_<version>.exe`.
2. If Windows shows **"Windows protected your PC"** (SmartScreen), click **More info** and then **Run anyway**. This message appears because the installer is not yet code-signed; it is expected.
3. Allow the administrator prompt (the installer needs administrator rights).
4. Accept the licence (MIT) and click **Next**.
5. Choose the install folder (the default `C:\Program Files\NI USB-6009 Logger` is fine).
6. **Components** (if shown):
 - *NI USB-6009 Logger application* — always installed.
 - *NI-DAQmx driver runtime* — keep it **ticked** if you have not installed the NI driver on this PC. It installs by itself without questions and can take several minutes.
7. **Additional tasks:**
 - *Create a desktop shortcut* — optional.
 - *Test the installation now* — leave **ticked**.
8. Click **Install**, wait, then **Finish**.

5.2 Read the installation test

If you left the test ticked, a box will report one of three results:

Message	Meaning / what to do
Driver OK and USB-6009 detected — everything works.	Done.
Driver OK. No DAQ device found yet.	Fine if the device isn't plugged in. Connect the USB-6009; the program finds it by itself.
The NI-DAQmx driver is not working yet.	Reboot the PC once, then start the program again. If it still fails, install NI-DAQmx from ni.com (see 5.3).

5.3 If the NI-DAQmx driver is missing

If your installer didn't include the driver, the program tells you at start-up and offers an **Open download page** button. Download and install **NI-DAQmx** from ni.com (search "NI-DAQmx download"), restart the PC if asked, then start the program again.

5.4 Installing the Python version (advanced users only)

Not needed if you use the installer. For the command-line tool or development, with Python 3.10+ and the NI-DAQmx driver already installed:

```
python -m venv .venv
.\.venv\Scripts\activate
pip install -e .[excel]          # command-line tool + Excel output
pip install -e .[gui,excel]     # also the graphical program (ni_usb6009_gui)
```

6. Starting the program for the first time

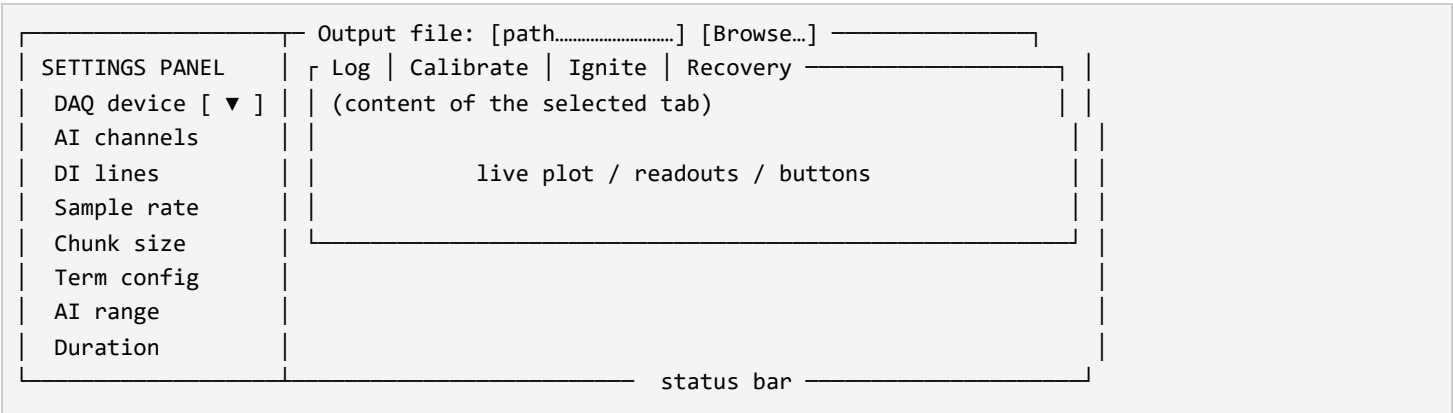
- 1. Plug in the USB-6009.
- 2. Start **NI USB-6009 Logger** from the Start menu or the desktop icon.
- 3. The window title shows the version. In the status bar you should read **"DAQ detected: Dev1 (USB-6009)"**.

If instead you see:

Status bar	Meaning
No DAQ detected — waiting for device...	The device isn't connected or isn't recognised yet. Check the cable; the program re-scans every 2 seconds, or press Refresh .
NI configuration service not running...	A Windows service of NI is stopped. See troubleshooting .

Your settings are remembered between sessions, so the second start looks like how you left it.

7. The main window



- **Left: settings panel** — settings that apply to every mode.
- **Top right: Output file** — where the data is saved; shared by *Log* and *Ignite*.
- **Tabs:** *Log*, *Calibrate*, *Ignite*, *Recovery*.

- **Status bar** (bottom): device status, then during a run the number of samples, elapsed time and data rate.
- A **log pane** repeats the program's messages.

8. Settings that apply to every mode

Field	What to enter	Default
DAQ device	Choose from the list. It is filled in automatically. You can type a name if the device isn't listed. Refresh rescans.	Dev1
AI channels	Analog channels to measure, comma-separated: ai0 or ai0,ai1,ai2. At least one is required.	ai0
DI lines	Optional digital inputs. port0/line0:7 (a whole range) or port0/line0,port0/line3. Leave empty for none. Only port0/line0-7 and port1/line0-3 exist. A name like D0 is refused.	empty
Sample rate	Samples per second per channel , 1–48 000 Hz.	1000 Hz
Chunk size	Samples read from the device in one go. Also how often the digital inputs are read: once per chunk.	1000
Term config	RSE (single-ended) or DIFF (differential). See 4.3.	RSE
AI range	Expected minimum and maximum voltage, –10 V ... +10 V. Also sets the plot's vertical axis.	–10 to +10 V
Duration	Run time in seconds. 0 shows "run until Stop".	run until Stop

Important limit — total speed. The USB-6009 can sample at most **48 000 samples per second *in total***, shared by all channels. So the maximum sample rate is $48\,000 \div \text{number of channels}$ (4 channels → 12 000 Hz; 8 channels → 6 000 Hz). The program checks this before starting and tells you the maximum.

Digital sampling speed. Digital inputs are read once per chunk. At 1000 Hz with a chunk of 1000 you get a digital reading once per second; with a chunk of 100, ten times per second. Smaller chunks make the program do more reads, but there is no other cost.

All settings are saved when you start a run and when you close the program.

9. Logging data (Log tab)

9.1 Steps

1. Fill in the settings panel (chapter 8).
2. Click **Browse...** next to *Output file* and choose where to save. The suggested place is Documents\NI6009 Logs\ni_<device>_<date>_<time>.csv. Choose **CSV** or **Excel (.xlsx)** with the file type, or by the extension you type.
3. **Start** is enabled only once an output file is chosen and a device is available. Click **Start**.
4. Watch the live plot (chapter 12) and the status bar.
5. The run ends when the *Duration* has passed, or when you click **Stop**.
6. A summary box appears with an **Open folder** button that takes you to your files.

9.2 Existing files are never overwritten

If the name you chose already exists, `_1`, `_2`, ... is added to the new file name. The name shown in the *Output file* box after you choose is the exact name that will be written.

9.3 What is in the file

One row per sample instant, with these columns:

Column	Meaning
timestamp_iso	Date and time when the sample was acquired (not when it was read from the device).
sample_index	Counter of the sample, starting at 0.
One column per AI channel	Voltage in volts (ai0, ai1, ...).
One column per DI line	0 or 1 (port0/line0, ...). Same value repeated within a chunk.

CSV opens in Excel or any text editor. **XLSX** is a native Excel workbook; note that an XLSX file only reaches the disk when the run *ends* (the always-flushed [recovery copy](#) protects you meanwhile).

10. Calibrating (Calibrate tab)

Use this to look at stable, averaged values — for instance to zero a load cell or check a sensor — **without** saving anything. No output file is needed.

Settings on the tab:

Field	Meaning	Default
Moving-average window	Seconds of data averaged for each shown value. 0 disables averaging.	5 s
Screen output rate	How many times per second the readout refreshes.	1 Hz
Internal sample rate	How fast the hardware samples to feed the average.	100 Hz

1. Set channels and terminal config in the settings panel.
2. Click **Start calibration**.
3. The large readout shows the moving average for each channel; the plot shows the raw signals.
4. Click **Stop** when finished.

The average needs one full window to settle: with a 5 s window, wait about 5 s after a change before reading the value.

11. Igniting (Ignite tab)

⚠ **Safety-critical function.** Read chapters 3 and 4.5 first. Do a complete dry run with a dummy load and **without** an igniter in the motor before real use.

11.1 Settings

Field	Meaning	Default
Buzzer DO line	Digital output driving the buzzer.	port1/line1
Igniter relay DO line	Digital output driving the relay.	port1/line0
Current-sense AI	Analog input measuring the voltage across the shunt.	ai2
Sense term config	RSE or DIFF for that channel (must match your wiring).	RSE
Shunt resistance	Value of your shunt, in ohms.	1.0 Ω
Continuity minimum	Minimum bias current that proves the igniter is connected.	0.2 mA
Leak maximum	If more current than this flows <i>before</i> firing, the system is blocked.	5 mA
Fire-confirm minimum	Current that must be seen <i>during</i> the pulse to confirm ignition.	300 mA
Buzzer warning time	How long the warning buzzer sounds before logging starts.	15 s
Stabilize time	Time between logging start and FIRE becoming available.	1 s
Relay pulse time	How long the relay stays on.	1 s

Also needed:

- An **output file** (ignition always logs).
- The **current-sense channel in the "AI channels" list** (e.g. ai0, ai2). The USB-6009 can run only one analog acquisition at a time, so the confirmation current is read from the same running acquisition. If you leave it out the pulse still happens, but the confirmation reading may be reported as an error.
- Buzzer and relay lines must be real output lines (port0/line0-7 or port1/line0-3) and different from each other.

11.2 The safety panel

Four LEDs: **DO lines LOW**, **Continuity**, **No leak current**, **Igniter relay**; a status line (with the live current in mA while arming); a bar showing how long you have held FIRE; and the buttons **ARM**, **FIRE**, **ABORT**.

11.3 The sequence, step by step

Step	What you do	What the system does
1. ARM	Click ARM and confirm the dialog ("buzzer will sound... relay stays OFF").	Starts the session. Forces both output lines LOW first.
2. Arming	Wait (you can ABORT any time).	The buzzer sounds for the warning time while the current is measured and shown live.
3. Check	—	Pass: continuity seen and no leak → continues. Fail: buzzer off and the message " <i>INHIBITED by safety failsafe</i> " with the reason (leak current too high, or no continuity). FIRE is never offered and the relay is never energised.
4. Logging	Wait.	Logging starts. After the stabilize time FIRE is enabled .
5. FIRE	Press and hold FIRE for 2 seconds (the bar fills; letting go early cancels).	The relay is switched on for the pulse time, the current is checked against <i>Fire-confirm minimum</i> , then the relay is switched off.

Step	What you do	What the system does
6. ABORT	Click ABORT at any moment.	Stops everything; output lines go LOW.

Rules built into the system:

- FIRE is never available before the system is ready, and is disabled again after the pulse, after an inhibit, or when the session ends.
- The current limits are checked by the program core, not by the buttons — the same protection as the command-line version.
- If the current during the pulse stays below *Fire-confirm minimum*, a warning asks you to check wiring, supply and igniter. The relay still finishes its pulse and switches off.
- A strongly **negative** current (shunt wired backwards) blocks arming with an explicit message.
- **Closing the window = ABORT.** A "stopping safely" box stays until the output lines are confirmed LOW, then the program closes.
- Unplugging the USB cable during a run stops the session; the outputs go LOW and the data written so far is kept.

11.4 Suggested dry run

1. Wire everything with a dummy load (e.g. a suitable resistor or lamp) instead of an igniter, or no ignition supply at all, and the igniter not in the motor.
2. Set a short warning time (e.g. 5 s) for the test.
3. ARM → confirm → check the LEDs: continuity lights up, leak LED stays OK.
4. Disconnect the dummy load and ARM again → expect *INHIBITED (no continuity)*.
5. Reconnect, ARM, wait for FIRE, hold it → hear/see the relay click for 1 s, check the logged current.
6. Press ABORT during another arming → confirm everything goes quiet.

12. The live plot

- One coloured line per AI channel with a legend; X axis = seconds since the start of the run, Y axis = volts (your *AI range*).
- It shows roughly the last 60 seconds; memory use stays constant however long you run. The **file** contains *all* data.
- **view:** drop-down at the top right: **follow live** (moves with new data) or **pause view** (freezes the picture so you can zoom and pan; the recording continues).
- Zoom with the mouse wheel; drag to pan (while paused).

13. Output files and the recovery copy

13.1 Recovery copy

For every logging or ignition run the program writes a second, parallel file:

```
<your output folder>\recovery\<your file name>_recovery.csv
```

- It is always a CSV, saved to disk after every chunk (an XLSX only reaches the disk at the end).

- When the run finishes normally it is renamed to ..._recovery_OK.csv.
- After a crash, an unplugged cable, an abort or any error, there is **no**_OK in the name — the partial data is safe and ready to be recovered.

13.2 The Recovery tab

Lists the recovery files (newest first) with their status (**OK** or **INTERRUPTED**) and size, from the current output folder's recovery subfolder and from the default logs folder. It refreshes at start-up, when you open the tab and after every run.

- **Refresh** — rescan.
- **Copy to...** — copy the selected file to a place you choose. If copying fails you get a message with the reason.

13.3 Where is everything?

What	Where
Default data folder	Documents\NI6009 Logs
Recovery copies	recovery\ subfolder next to your output file
Your settings	Windows registry HKCU\Software\steeman.be\NI USB-6009 Logger (managed by the program)

14. Using the command-line version

For scripting, open **PowerShell** (with the Python install active, chapter 5.4) and run `ni_usb6009_logger`. A channel list is the only required option.

```
ni_usb6009_logger --help
```

14.1 Examples

```
# Log ai0 at 1000 Hz, auto-named file under .\logs, print first 10 rows
ni_usb6009_logger --device Dev1 --channels ai0 --rate 1000 --term RSE --print-first 10

# Two analog + four digital lines, custom CSV
ni_usb6009_logger --device Dev1 --channels ai0,ai1 --digital port0/line0:3 `
  --rate 100 --outfile .\logs\run.csv

# Excel output, 30 seconds, progress bar
ni_usb6009_logger --device Dev1 --channels ai0,ai1 --rate 500 `
  --outfile .\logs\run.xlsx --duration 30 --progress bar

# Differential
ni_usb6009_logger --device Dev1 --channels ai0,ai1 --rate 1000 --term DIFF

# Calibration (screen only, 5 s moving average)
ni_usb6009_logger --device Dev1 --channels ai0,ai1 --calibrate --calib-window 5 `
  --calib-sample-rate 100 --rate 1

# Ignition with buzzer, relay and current-sense failsafe
ni_usb6009_logger --device Dev1 --channels ai0,ai2 --rate 1000 --term RSE `
  --ignite --buzzer-line port1/line1 --igniter-line port1/line0 `
  --igniter-sense-ai ai2 --sense-term RSE --shunt-ohms 1.0 `
  --continuity-min-ma 0.2 --leak-max-ma 5 --fire-confirm-ma 300
```

(In cmd.exe, use ^ instead of the backtick for line continuation.)

14.2 Options

Option	Meaning	Default
--device	Device name (as in NI MAX)	Dev1
--channels	AI channels, e.g. ai0,ai1 (required)	—
--digital	DI lines, e.g. port0/line0:7	none
--rate	Sample rate (Hz)	1000 (1 in calibration)
--chunk	Samples per read	1000
--vmin, --vmax	Expected voltage range	−10, 10
--term	RSE or DIFF (NRSE is accepted but refused by this device)	RSE
--outfile	Output path; auto-named in .\logs if omitted	—
--format	csv or xlsx (otherwise from the extension)	csv
--duration	Seconds; omit to run until Ctrl+C	—
--progress	auto, none, counter, bar	auto
--update-interval	Seconds between progress updates	0.5
--print-first	Print the first N rows	0
--debug	Extra diagnostic output	off

Option	Meaning	Default
--calibrate	Calibration mode	off
--calib-window	Moving-average window (s, 0 = off)	5
--calib-show-raw	Also print raw values	off
--calib-sample-rate	Internal rate in calibration (Hz)	100
--ignite	Enable ignition sequence	off
--buzzer-line, --igniter-line	Output lines (required with --ignite)	—
--arm-seconds	Buzzer warning time	15
--stabilize-seconds	Wait after logging starts	1
--pulse-seconds	Relay on-time	1
--igniter-sense-ai	Channel measuring the shunt, e.g. ai2	—
--shunt-ohms	Shunt value	1.0
--continuity-min-ma / --leak-max-ma / --fire-confirm-ma	Current limits	0.2 / 5 / 300
--sense-rate	Sense sample rate (Hz)	100
--sense-term	Terminal config for the sense channel	DIFF

Note: the command-line default for --sense-term is **DIFF**, while the GUI shows RSE; set it explicitly to match your wiring.

14.3 Stopping, exit codes

- **Ctrl+C** once = clean stop (current chunk finishes, files close, outputs LOW). A **second** Ctrl+C aborts immediately (exit code 130) if the device is stuck.
- Exit code **2** = a setting was refused before starting (with an explanation).

15. Troubleshooting

Symptom	Cause / fix
Program says the NI-DAQmx driver is missing and exits	Click <i>Open download page</i> , install NI-DAQmx, restart the PC, start the program again.
"No DAQ detected — waiting for device..."	Check the USB cable and port; try another port. Press Refresh . Make sure the device appears in NI MAX. You can also type Dev1 in the device field.
"NI configuration service not running" or an error mentioning <i>MIG</i>	NI background services are stopped. Reboot the PC. Otherwise, in an administrator PowerShell: Get-Service *NI* Sort-Object Status, Name then Start-Service -Name "nidevldu", "mxssvr", "nimDNSResponder"; or restart the <i>NI Configuration Manager</i> service in Windows Services.
Start / ARM is greyed out	No device available, or (for Log/Ignite) no output file chosen yet.
"Cannot start" dialog	A setting is invalid; the message names the field. Typical: rate × channels above 48 000; DIFF on ai4–ai7; wrong digital line name; AI minimum above maximum; empty channel list.

Symptom	Cause / fix
Digital line D0 refused	Use the full name: port0/line0.
Run stopped, "check USB cable" dialog	The device disconnected. Data written so far is in the output <i>and</i> recovery files named in the dialog. Re-plug and start a new run.
Arming says "INHIBITED — no continuity"	Igniter not connected, bias resistor missing, sense wire off, or <i>Continuity minimum</i> set too high.
"INHIBITED — leak current"	Current flows before firing: short, wet/dirty connection, wrong wiring, relay stuck. Fix before retrying; do not raise the limit just to get past it.
Reversed-current message	The shunt or the AI wires are swapped.
Fire-confirm warning	Current during the pulse stayed below the limit: check supply voltage, fuse, igniter, shunt value and wiring.
Values look noisy or drift	Use DIFF wiring, shorter/shielded wires, a common ground; use Calibrate with a longer averaging window.
Excel file is missing after a crash	XLSX is only written at the end. Use the Recovery tab and copy the ..._recovery.csv.
Windows SmartScreen warning at install	Click <i>More info</i> → <i>Run anyway</i> .

16. Uninstalling

Windows Settings → Apps → NI USB-6009 Logger → Uninstall, or Start menu → NI USB-6009 Logger → Uninstall NI USB-6009 Logger. Your data files in Documents\NI6009 Logs are **not** deleted. The NI-DAQmx driver is a separate product and stays installed (remove it separately from Apps if you no longer need it).

17. Technical reference and limits

Item	Value
Analog inputs	8 single-ended (RSE) or 4 differential (DIFF, ai0–ai3), ± 10 V max
Maximum sampling	48 000 S/s total, shared by all channels
Digital lines	port0/line0–7, port1/line0–3; static (not clocked)
Terminal modes	RSE, DIFF only (no NRSE)
Digital snapshot	Once per chunk
Timestamps	Back-dated to when each sample was acquired
File formats	CSV, XLSX
Overwrite protection	Always; _1, _2... suffix
Recovery	CSV, flushed every chunk; _OK on clean finish
Live plot memory	Bounded; ~60 s window
Ignition timing	The AI stream is read in ~20 ms sub-blocks so the relay opens on time and ABORT acts within a sub-block
Operating system	Windows 10/11 64-bit

Item	Value
Not bundled	NI driver files (installed separately by NI-DAQmx)

Testing without hardware: setting the environment variable `NI_USB6009_FAKE=1` starts the program with a built-in fake device (no driver needed); useful for training and demos. It is more forgiving than the real device. For a more realistic test, create a *simulated USB-6009* in NI MAX (*Devices and Interfaces* → *Create New* → *NI-DAQmx Simulated Device*).

Self-test: `NI6009Logger.exe --selftest` (from the install folder) prints one line and returns 0 (driver and device OK), 1 (driver OK, no device) or 2 (driver missing or NI service not running).